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Deep learning-based intrusion detection for IoT networks: a scalable and efficient approach

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Abstract
The rapid expansion of the Internet of Things (IoT) has revolutionized industries by enabling seamless connectivity, but it has also introduced significant security vulnerabilities, making IoT networks prime targets for cyberattacks. Traditional intrusion detection systems often struggle to cope with the high volume and dynamic nature of IoT traffic, necessitating the development of more robust and intelligent security mechanisms. This research presents a deep learning-based approach for real-time threat detection in IoT networks, leveraging advanced models such as 1D convolutional neural networks (CNNs), long short-term memory (LSTM) networks, recurrent neural networks (RNNs), and multi-layer perceptrons (MLPs) to enhance intrusion detection. The study utilizes the CIC IoT-DIAD 2024 dataset, a comprehensive collection of flow-based network traffic containing both benign and attack scenarios. The proposed models were trained and evaluated on flow-based feature sets, optimizing hyperparameters to maximize accuracy, recall, and F1-score. In multi-class classification, 1D CNN achieved the highest accuracy of 99.12%, followed by LSTM (98.98%), RNN (98.43%), and MLP (97.21%). For binary anomaly detection, 1D CNN again demonstrated superior performance with an accuracy of 99.53%, while LSTM, RNN, and MLP achieved 99.52%, 99.25%, and 98.78%, respectively. The results indicate that 1D CNN is the most effective model for real-time IoT intrusion detection, excelling in feature extraction and attack classification. The findings contribute to the development of scalable and efficient deep learning-based security solutions, improving the ability to detect and mitigate cyber threats in IoT environments.

Keywords IoT security, Intrusion detection system (IDS), Deep learning in cybersecurity, Flow-based network analysis, Anomaly detection in IoT

1 Introduction

IoT intrusions refer to unauthorized and malicious activities targeting IoT networks, exploiting vulnerabilities in device communication, authentication mechanisms, and network protocols to compromise security, disrupt services, or gain unauthorized access [1, 2]. These intrusions manifest in various attack forms, including Brute Force attacks, where attackers systematically attempt multiple credential combinations to gain unauthorized access to IoT devices. Distributed denial-of-service (DDoS) and denial-of-service (DoS) attacks flood network resources with excessive traffic, rendering IoT devices or services unavailable [3, 4]. Mirai botnet attacks leverage infected IoT devices to launch large-scale cyberattacks, often overwhelming targets with massive traffic volumes. Spoofing attacks manipulate network identity by forging IP or MAC addresses, deceiving IoT systems into granting illegitimate access [5, 6]. SQL Injection (SQLi) attacks exploit vulnerabilities in database-driven applications, injecting malicious SQL queries to manipulate or extract sensitive data from IoT-enabled services. Cross-Site Scripting (XSS) attacks inject malicious scripts into web-based IoT interfaces, compromising user sessions and

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enabling unauthorized control over connected devices [7–9].

Addressing IoT security threats through advanced research is essential to safeguarding critical infrastructures, smart environments, and personal devices from increasingly sophisticated cyberattacks [10]. IoT networks are deeply embedded in healthcare systems, industrial automation, smart cities, and financial services, making them attractive targets for attackers seeking to disrupt services, steal sensitive data, or take control of connected devices. The consequences of undetected intrusions can be severe, ranging from service downtime and financial losses to privacy breaches and safety hazards in mission-critical applications [11, 12]. Traditional security measures, including signature-based and rule-based intrusion detection systems, are inadequate against zero-day exploits, evolving malware, and large-scale botnet attacks like Mirai. Without an intelligent, adaptive, and real-time threat detection system, IoT networks remain vulnerable to invisible and persistent attacks that can go undetected for long periods. Researching and developing deep learning-based intrusion detection systems enables faster, more accurate, and autonomous threat detection, reducing false alarms and improving network resilience against emerging cyber threats. This advancement will not only strengthen IoT security but also enhance trust in IoT deployments, paving the way for safer and more scalable adoption of connected technologies in modern society [13, 14].

Detecting and mitigating IoT intrusions is a complex challenge due to the heterogeneous nature, large-scale deployment, and resource constraints of IoT devices. Unlike traditional computing environments, IoT networks consist of diverse devices with varying hardware capabilities, operating systems, and communication protocols, making it difficult to implement a standardized security framework. The high volume and dynamic nature of IoT network traffic further complicate intrusion detection, as distinguishing between benign anomalies and malicious activities requires context-aware intelligence and adaptive detection mechanisms [15, 16]. Additionally, many IoT devices operate with minimal computational power and storage, restricting the deployment of heavyweight security solutions. Conventional signature-based detection methods fail to keep up with zero-day attacks, polymorphic malware, and evolving cyber threats, leading to high false positive and false negative rates [17, 18]. Furthermore, attackers continuously develop new evasion techniques, such as encrypted command-and-control traffic, making it challenging to detect threats using traditional network monitoring tools. The sheer scale of IoT ecosystems, with millions of connected devices generating massive amounts of data, demands real-time, high-performance intrusion detection systems that can analyze patterns efficiently without overwhelming computational resources [19, 20]. These complexities make IoT security a multi-dimensional challenge, requiring advanced deep learning techniques that can learn, adapt, and respond to emerging threats in real time.

To address the challenges of IoT intrusion detection, this research develops a deep learning-based intrusion detection system (IDS) that leverages flow-based network traffic analysis to detect and classify cyber threats in real-time. The study employs state-of-the-art deep learning models, including convolutional neural networks (CNNs), long short-term memory (LSTM) networks, recurrent neural networks (RNNs), and multi-layer perceptrons (MLPs), to analyze IoT traffic patterns and distinguish between benign and malicious activities. Using the CIC IoT-DIAD 2024 dataset, which contains comprehensive flow-based features from multiple attack categories, the models are trained to recognize complex attack signatures and behavioral anomalies. The research focuses on two core tasks: multi-class attack classification, which identifies specific types of cyber threats, and binary anomaly detection, which differentiates between normal and malicious traffic. By integrating feature extraction, data preprocessing, and model optimization, this study ensures high detection accuracy while maintaining computational efficiency, making it suitable for deployment in resource-constrained IoT environments. The proposed approach enhances real-time threat detection capabilities, reduces false alarms, and improves overall IoT security by providing a scalable, intelligent, and adaptive intrusion detection framework.

Traditional machine learning techniques, such as decision trees, support vector machines, and k-nearest neighbors, have been widely used for intrusion detection in IoT networks. However, these models often require extensive feature engineering and struggle with high-dimensional and sequential data. Deep learning methods, particularly CNNs, LSTMs, RNNs, and MLPs, have demonstrated superior capability in automatically extracting hierarchical features, modeling temporal dependencies, and improving classification accuracy. CNNs effectively capture spatial relationships in network traffic, while LSTMs and RNNs excel in processing sequential data to detect time-based attack patterns. MLPs, on the other hand, provide a dense and fully connected structure, ensuring robustness in classification. Unlike hybrid approaches that often involve complex model integration and higher computational overhead, our selected deep learning models offer a balance between efficiency and accuracy, making them well-suited for real-time intrusion detection in IoT environments. Additionally, previous studies often focus solely on either anomaly detection or attack classification, whereas our approach integrates both binary anomaly detection and multi-class classification, ensuring a comprehensive intrusion detection framework. Furthermore, many existing solutions suffer from high false positive rates, causing inefficiencies in real-world applications. By optimizing feature selection, hyperparameters, and model architectures, this approach minimizes false alarms and enhances detection precision. These advantages position this deep learning-based IDS as a more robust, efficient, and adaptive security solution for protecting IoT networks against a wide range of cyber threats. The key contributions of this research are:

- Development of a deep learning-based intrusion detection system (IDS) for IoT networks.
- Comprehensive analysis using the CIC IoT-DIAD 2024 dataset which is a real-world, flow-based IoT intrusion dataset containing multiple attack categories.
- Designed a dual-level detection framework that enables both multi-class attack classification and binary anomaly detection for real-time threat identification.
- Reduced false positive rates and improved detection precision by selecting the most efficient deep learning model for IoT security.
- Provided a scalable and adaptive IDS that enhances real-time IoT security and serves as a foundation for future AI-driven intrusion detection research.

The next section of the paper reviews existing intrusion detection systems for IoT, including traditional rule-based methods, machine learning approaches, and deep learning models, highlighting their limitations in handling evolving cyber threats. "Methodology" section details the proposed deep learning-based intrusion detection framework, explaining the dataset, data preprocessing, feature selection, and the implementation of CNN, LSTM, RNN, and MLP models for multi-class attack classification and anomaly detection. "Experimental results and analysis" section presents the performance evaluation of each model for both binary anomaly detection and multi-class classification. "Conclusion" section summarizes the research contributions, emphasizing the effectiveness of CNN for IoT intrusion detection, the significance of deep learning in improving IoT security, and potential future enhancements for real-time deployment in IoT networks.

2 Related works

Intrusion detection systems (IDS) play a pivotal role in securing modern networks against malicious cyber threats. Over the past decade, traditional signature-based IDS solutions have struggled to keep pace with evolving attack vectors, leading to an increased reliance on artificial intelligence (AI) and deep learning (DL) techniques for robust network defense [21, 22]. This section explores recent advancements in deep learning-driven IDS, highlighting key methodologies, challenges, and future research directions.

Several early studies explored supervised machine learning for intrusion detection, employing classifiers such as Support Vector Machines (SVM), Decision Trees (DT), Random Forest (RF), Naïve Bayes (NB), and k-Nearest Neighbors (k-NN). For instance, SVM to detect network anomalies, reporting high detection rates but limited scalability due to computational complexity [23]. Similarly, utilizing RF and DT classifiers, achieving improved accuracy but suffering from high false positive rates when applied to imbalanced datasets. Bayes-based classifiers, while computationally efficient, struggled with complex attack patterns, leading to low recall for novel intrusions. To overcome feature selection challenges, ensemble learning techniques such as AdaBoost and XGBoost were introduced. Ensemble learning model combining DT, RF, AdaBoost, and CatBost, demonstrated improved attack detection [24–27]. However, these models required extensive feature engineering, making them less adaptable to rapidly evolving cyber threats. While ML-based techniques provided an improvement over traditional rule-based systems, their reliance on handcrafted features and inability to learn hierarchical representations limited their generalization capabilities [28].

Deep learning has revolutionized intrusion detection by automating feature extraction and capturing complex patterns in high-dimensional network traffic data. The study by Marwa Keshk et al. [24] proposes an explainable deep learning-based IDS using LSTM and interpretability techniques (SHAP, PFI, ICE, PDP) on NSL-KDD, UNSW-NB15, and TON_IoT datasets, achieving 99.02%, 98.76%, and 99.31% accuracy, respectively. However, it faces high computational complexity, limiting real-time deployment, vulnerability to adversarial attacks, and dataset dependency, restricting adaptability to evolving threats. While explainability enhances interpretability, it adds computational overhead, making it less scalable for real-time IoT security. Albara Awajan [29] presents a deep learning-based IDS for IoT environments. The study proposes a four-layer fully connected (FC) deep neural network designed to identify cyber threats, particularly Blackhole, DDoS, Opportunistic Service, Sinkhole, and Wormhole attacks. The system is protocol-independent, making it adaptable for various IoT infrastructures. The results show an average accuracy of 93.74% which is very low in the context of the present AI world. The lack of comparative analysis with existing IDS approaches raises concerns about the model's superiority over established methods. Additionally, the system's dependency on a dataset generated within the experiment suggests potential issues with generalizability. The model has not been evaluated on diverse intrusion datasets, limiting its applicability to real-world IoT scenarios.

Asgharzadeh et al. [30] proposed a hybrid IDS combining CNN, KNN, and the BMEGTO optimizer to enhance feature selection for IoT security, achieving 99.99% accuracy on NSL-KDD. However, the approach is computationally expensive, limiting real-time deployment. Additionally, its resilience against adversarial attacks and efficiency compared to lightweight IDS models remain unexplored. The study by Elnakib et al. [31] presents an Enhanced Intrusion Detection Model (EIDM) for IoT security, using deep learning for multi-class classification of network traffic. The model classifies 15 traffic behaviors, including 14 attack types, using the CICIDS2017 dataset. Comparative analysis with state-of-the-art IDS approaches demonstrates EIDM's superior classification accuracy of 95%. While EIDM outperforms previous existing models, improvements in lightweight architectures and enhanced explainability are required for practical IoT applications. Heidari et al. [32] introduces a blockchain-based intrusion detection system (IDS) for Internet of Drones (IoD) environments, leveraging Radial Basis Function Neural Networks (RBFNNs). The study highlights the integration of blockchain to enhance security, ensuring decentralized and tamper-proof data storage. The authors employ a transfer learning approach to optimize training efficiency. Despite achieving high detection accuracy, the model's adaptability to real-time drone networks and its computational overhead remain concerns. Gómez and Muñoz [33] focus on using deep learning techniques for detecting and classifying malware in Android devices. The study utilizes three distinct malware datasets and demonstrates the effectiveness of deep learning in malware detection, classifying applications into categories like Adware and Banking. While achieving high detection accuracy, the paper has limitations, including a lack of real-time deployment considerations and insufficient focus on adversarial attacks. It also does not explore how the model can adapt to the evolving nature of malware threats.

Madhu et al. [34] proposes a device-based intrusion detection system (DIDS) leveraging deep learning for IoT security. The study integrates AI techniques to enhance security and improve real-time detection in large-scale networks. The DIDS model predicts unknown attacks, reducing computational overhead and false alarm rates. The system is evaluated against standard algorithms, achieving an accuracy of 99%. However, the study primarily focuses on detection accuracy without addressing the real-time deployment challenges and adversarial robustness. Additionally, the paper lacks a detailed comparative analysis with other deep learning-based intrusion detection frameworks, leaving gaps in understanding its scalability and adaptability across diverse IoT environments. The study by Alghamdi and Bellaiche proposes an ensemble deep learning-based IDS for IoT using LSTM, CNN, and ANN within a Lambda architecture, achieving high accuracy (99.93%). However, the approach suffers from high computational overhead, increased inference time, and vulnerability to adversarial attacks. Additionally, reliance on pre-processed datasets limits real-world applicability, and scalability remains a challenge for large-scale IoT networks [35]. Muñoz et al. [36] examines vulnerabilities in various IoT communication protocols such as WiFi, Bluetooth, NFC, and GPS, this research strengthens the discussion on the generalizability of the proposed IDS across diverse IoT environments. The study emphasizes that IoT networks in real-world scenarios consist of heterogeneous communication protocols and architectures, which may lead to variations in attack patterns. Therefore, a comprehensive IDS must be able to adapt to these dynamic environments to provide robust security.

Despite significant advancements in machine learning (ML) and deep learning (DL)-based intrusion detection systems (IDS), several challenges persist in cybersecurity applications. Many intrusion datasets suffer from severe high false positive rate for class imbalance. Additionally, IDS models are vulnerable to adversarial attacks, where cybercriminals manipulate input data to evade detection, necessitating robust defense mechanisms to enhance model resilience. Another critical challenge is the balance between feature engineering and automation, while deep learning methods automate feature extraction, ensuring model interpretability and efficient feature selection remains a significant concern in cybersecurity applications. To address these limitations, this research proposes a 1D CNN-based intrusion detection system, using efficient feature extraction through convolutional operations, enabling real-time threat detection while maintaining computational efficiency and high accuracy with lower false positive rate. The proposed approach is highly scalable, making it adaptable to large-scale datasets while overcoming the shortcomings of traditional ML-based IDS. Furthermore, the robust performance across multiple datasets demonstrates its ability to detect a diverse range of cyber threats, ensuring a highly reliable and effective intrusion detection system.

3 Methodology

This section describes the methodology of the research, covering the complete pipeline from dataset preparation, data preprocessing, model development, and training to performance evaluation.

3.1 Dataset

The research utilizes the CIC IoT-DIAD 2024 dataset [37], a flow-based IoT intrusion detection dataset designed for both device identification and anomaly detection. It contains network traffic data from 105 IoT devices under both benign and malicious conditions, covering 33 different cyberattacks categorized into seven major attack types: DDoS, DoS, Brute Force, Spoofing, Web-based attacks, SQL Injection, and Mirai botnet attacks. The dataset is structured into two subdirectories: flow-based features (extracted using CICFlowMeter) for anomaly detection and attack classification, and packet-based features for device identification. The dataset comprises 84 distinct features, capturing various network flow characteristics essential for intrusion detection. These features include packet-based attributes (e.g., flow duration, packet size), traffic-based attributes (e.g., flow byte rate, inter-arrival time), and protocol-related attributes (e.g., TCP flag counts, source/destination ports). Each record in the dataset contains detailed flow-level attributes, including packet statistics, time-based metrics, header flags, and protocol-specific features, enabling deep learning models to learn attack patterns effectively. The dataset provides a well-labeled ground truth, allowing for a supervised learning approach to intrusion detection. By leveraging real-world IoT traffic, this dataset ensures the reliability and applicability of the proposed intrusion detection models for scalable and real-time IoT security solutions. The major attack scenarios considered in this research are mentioned in Fig. 1.

3.2 Dataset preprocessing

To ensure the dataset is clean, structured, and suitable for deep learning models, a systematic preprocessing pipeline is applied. This step is crucial for improving the performance of the models by handling missing values, normalizing numerical features, encoding categorical attributes, and ensuring data consistency [38]. The dataset preprocessing is performed in the following stages:

The dataset contains numerical attributes extracted from IoT network traffic, and some of these values are infinite or extremely large due to network anomalies or computational errors. These values are replaced with NaN (Not a Number) and subsequently removed to prevent instability in training. Removing missing and infinite values ensures that models learn from accurate and clean data, reducing training instability. The operation is represented as

$$X_{cleaned} = X \setminus \{x \in X | x = \infty or x = -\infty\} \quad (1)$$

where, X represents the dataset, and X_{cleaned} is the resulting dataset after removing infinite values. To ensure model reliability, all rows containing NaN values are removed mention in Eq. 2. This prevents potential biases in learning due to missing or erroneous values.

$$X_{final} = X_{cleaned} \setminus \{x \in X_{cleaned} | x = NaN\} \quad (2)$$

where X_{final} is the dataset with only complete records, ensuring high-quality training data.

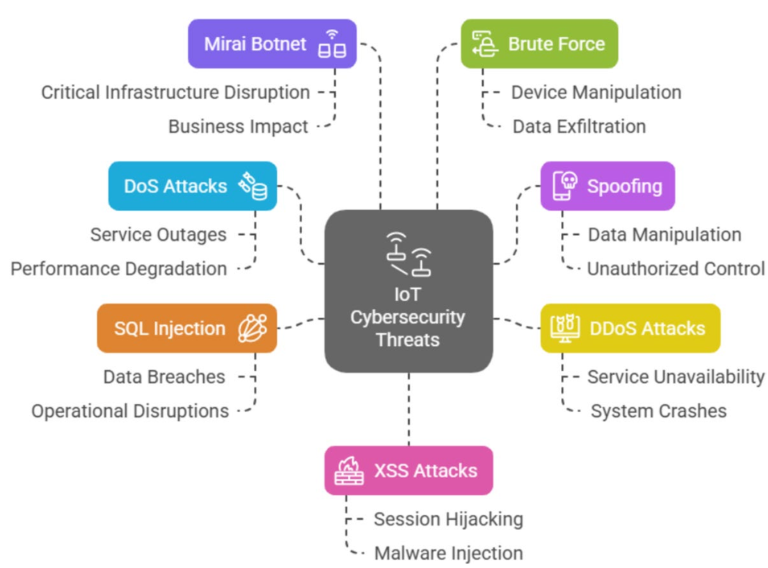

Fig. 1 Attack scenarios of the CIC IoT-DIAD dataset

The dataset contains both numerical and categorical features. To ensure consistency across different numerical attributes, StandardScaler is used to normalize numerical features. Each feature x is transformed as follows:

$$x\prime = \frac{x - \mu}{\sigma} \tag{3}$$

where μ is the mean and σ is the standard deviation of the feature. This ensures that all numerical features have zero mean and unit variance, improving model convergence during training. Normalization also allows deep learning models to converge faster by preventing features with large magnitudes from dominating learning.

Categorical features are converted into numerical representations using Label Encoding, which assigns a unique integer value to each category. Given a categorical feature C with unique values $\{c_1, c_2, \ldots, c_n\}$, label encoding assigns:

$$C\prime = f(C) = \{0, 1, \ldots, n - 1\} \tag{4}$$

where $f(C)$ is the encoding function that maps categorical values to integers. Encoding categorical features enables models to understand protocol-specific attributes, improving classification accuracy.

The dataset is split into training (80%) and testing (20%) subsets to evaluate model performance. This ensures that the models are trained on a majority of the data while reserving a portion for evaluation and validation. To maintain consistency between training and testing data, the previously fitted StandardScaler is applied to scale the test set using parameters learned from the training set:

$$X_{train}\prime = StandardScaler(X_{train}) \tag{5}$$

$$X_{test}\prime = StandardScaler(X_{test}) \tag{6}$$

where, $X\prime$ represents the transformed dataset.

Through these preprocessing techniques, the dataset becomes optimized for deep learning-based intrusion detection models, enhancing the accuracy, efficiency, and generalization of the proposed security framework.

The amount of data for each class of the CIC IoT-DIAD dataset is presented in Fig. 2. After preprocessing, the dataset is reshaped to match the input requirements of different deep learning models.

In real-world network environments, traffic flow is inherently imbalanced, with certain types of attacks occurring more frequently than others. Instead of artificially balancing the dataset using oversampling or data augmentation techniques, the proposed approach preserves the natural distribution of network traffic, ensuring that the model learns from real-world attack patterns. Deep learning architectures like 1D CNN, LSTM, RNN, and MLP have the ability to extract hierarchical and temporal features, allowing them to generalize well even in the presence of imbalanced data.

3.3 Model development

This section describes the model development process, including the architecture design, selection of

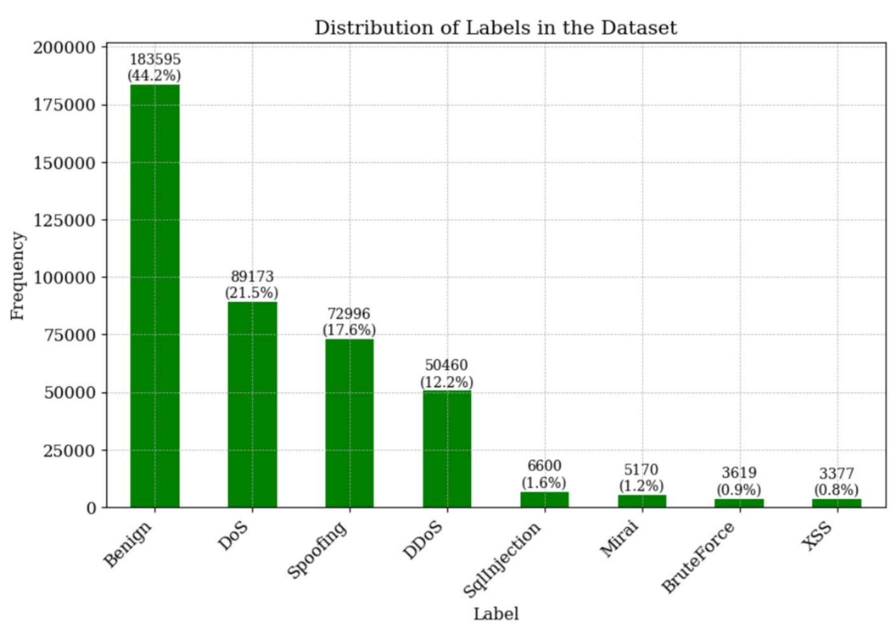

Fig. 2 Amount of data for each class after preprocessing

hyperparameters, and implementation of deep learning-based intrusion detection models for IoT security.

The 1D Convolutional Neural Network (1D CNN) is employed in this research to efficiently detect intrusions in IoT networks by extracting meaningful spatial and temporal patterns from network traffic data. CNNs are well-suited for feature extraction due to their ability to capture local dependencies and hierarchical patterns within sequential data. The proposed 1D CNN model is designed for both multi-class classification, distinguishing between different types of cyber threats, and binary classification, detecting anomalous activities. The architecture of 1D CNN for this research is mentioned in Fig. 3.

The 1D CNN model follows a sequential architecture consisting of multiple convolutional layers, pooling layers, and fully connected layers. The network begins with a series of 1D convolutional layers that apply filters over network traffic features to extract relevant patterns. These layers use the ReLU (Rectified Linear Unit) activation function to introduce non-linearity and enhance the model's ability to learn complex patterns. The convolutional layers are followed by MaxPooling layers, which reduce the dimensionality of extracted features while retaining critical information, thereby improving computational efficiency and preventing overfitting.

After feature extraction, the Flatten layer converts the multi-dimensional feature maps into a 1D vector, which is then fed into fully connected dense layers. The final output layer utilizes the softmax activation function for multi-class classification, enabling the model to predict probabilities for each intrusion category. In the case of binary classification (anomaly detection), the output layer is modified with a sigmoid activation function, which provides a probability score for classifying network traffic as either benign or an attack.

The capability of 1D CNN to analyze structured network traffic data makes them highly effective for intrusion detection. Unlike sequential models like LSTMs and RNNs, which rely on temporal dependencies, 1D CNNs efficiently extract spatial and hierarchical features from packet-based data, enabling rapid detection of attack patterns. The convolutional layers help in identifying critical intrusion signatures, while pooling operations enhance computational efficiency. This structure makes 1D CNNs particularly advantageous for real-time IoT security applications.

The performance of the CNN model is optimized using a carefully selected set of hyperparameters to ensure efficient feature extraction and accurate intrusion detection. The model employs 3-sized filters in each convolutional layer to capture fine-grained patterns from network traffic data, with 64, 128, and 256 filters in successive layers to enable multi-scale feature extraction. To enhance computational efficiency and mitigate overfitting, MaxPooling layers with a pool size of 2 are applied after each convolutional layer for dimensionality reduction. The ReLU activation function is utilized in convolutional and dense layers to improve gradient propagation and accelerate convergence. The model is trained using the Nadam (Nesterov-accelerated Adaptive Moment Estimation) optimizer, which dynamically adjusts learning rates to enhance training efficiency. The loss function is selected based on the task: categorical cross-entropy for multi-class classification and binary cross-entropy for anomaly detection. Training is conducted using a batch size of 32 for up to 200 epochs, with early stopping implemented to prevent overfitting and ensure optimal performance.

The long short-term memory (LSTM) model is employed in this research to effectively capture the temporal dependencies in network traffic data, which is crucial for intrusion detection. Unlike traditional neural networks, LSTMs are designed to handle sequential data by maintaining long-term dependencies through gated memory cells, making them highly suitable for analyzing time-series network traffic patterns. The architecture is displayed in Fig. 4.

The model is structured with multiple LSTM layers to extract meaningful temporal representations of network behavior. The LSTM model is trained for both multi-class intrusion detection and binary anomaly detection, effectively distinguishing between different types of attacks and normal traffic.

The LSTM model architecture is designed to capture complex attack patterns in network traffic through a stacked structure of four LSTM layers with 256, 128,

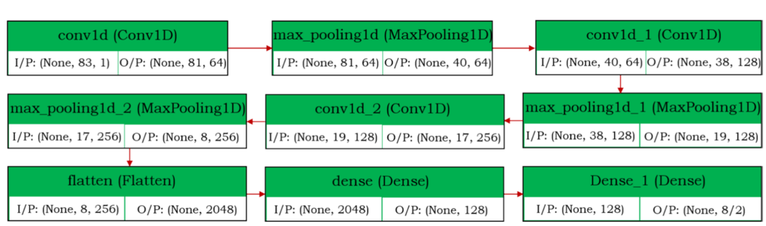

Fig. 3 Architecture of 1D CNN

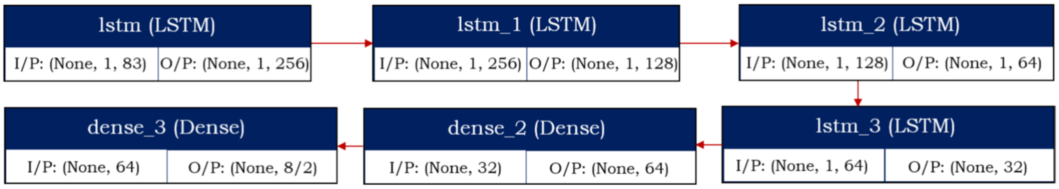

Fig. 4 Architecture of LSTM

64, and 32 units, respectively. The first three LSTM layers return sequences, enabling deeper feature extraction, while the final LSTM layer does not return sequences, ensuring a refined feature representation. A Dense layer with 64 neurons is included to further enhance the feature learning process, followed by an output layer with a softmax activation function for multi-class classification and binary cross-entropy for anomaly detection. The model is optimized with tanh activation in LSTM layers to regulate output values and mitigate gradient vanishing issues. The Nadam optimizer is employed for efficient learning by dynamically adjusting the learning rates. The loss function is set to categorical cross-entropy for multi-class classification and binary cross-entropy for anomaly detection. The model is trained using a batch size of 32 for up to 200 epochs, with early stopping applied to prevent overfitting by monitoring validation loss, ensuring robustness and optimal performance.

The LSTM model excels in learning sequential dependencies, making it highly effective in detecting sophisticated network intrusions such as DDoS, SQL Injection, Spoofing, and BruteForce attacks. The sequential nature of the model allows it to track behavioral changes over time, significantly improving the detection accuracy. Compared to traditional models, LSTM provides superior performance in identifying slow and evolving attack patterns by learning the sequence of network events.

The MLP model is employed as a fully connected deep learning architecture for detecting multi-class IoT intrusions. It operates by learning hierarchical representations of network traffic features through a series of fully connected dense layers. The model consists of four hidden layers with 256, 128, 64, and 32 neurons, respectively, each activated by ReLU to introduce non-linearity and improve feature extraction. To mitigate overfitting, Dropout layers (0.5 probability) are incorporated after each hidden layer, preventing co-adaptation of neurons and enhancing generalization. The final output layer uses softmax activation, ensuring probabilistic classification across the eight intrusion classes in the dataset. The architecture of utilized MLP is shown in Fig. 5.

The MLP model is trained using Nadam optimizer, which adapts learning rates dynamically to enhance convergence speed and stability. The loss function is categorical cross-entropy for multi-class classification, ensuring effective probability distribution learning. Training is conducted using a batch size of 32, and the model runs for up to 200 epochs, with early stopping applied based on validation loss to prevent overfitting. Through this architecture, the MLP model efficiently captures attack patterns in IoT networks, demonstrating high accuracy, precision, recall, and F1-score, making it a viable approach for intrusion detection.

The recurrent neural network (RNN) model is implemented to capture sequential dependencies in network traffic data, making it well-suited for IoT intrusion detection. Unlike traditional feedforward networks, RNNs maintain hidden states, allowing them to retain contextual information across time steps. The proposed RNN architecture consists of four stacked SimpleRNN

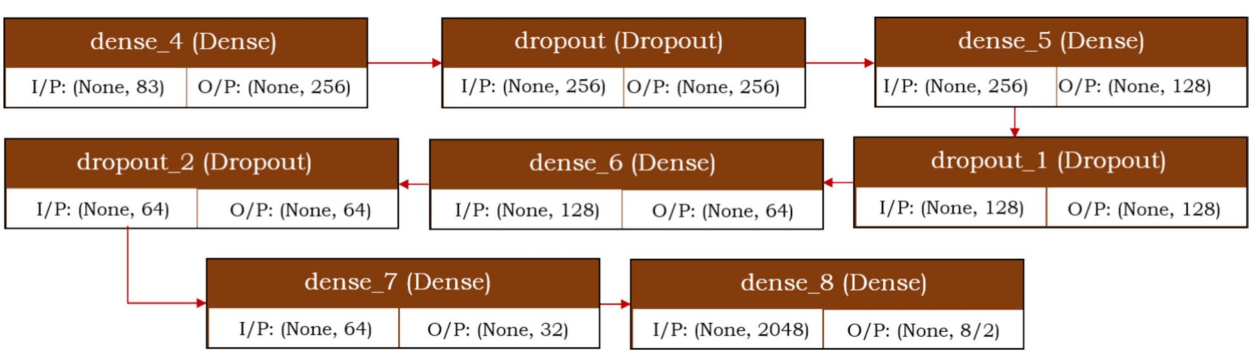

Fig. 5 Architecture of MLP

layers, each with 256, 128, 64, and 32 units, respectively. The first three layers return sequences, enabling hierarchical feature extraction, while the final RNN layer does not return sequences, feeding the extracted features into a dense layer with 64 neurons. The final output layer is activated by softmax for multi-class classification, distinguishing between different attack types, while binary cross-entropy loss is employed for anomaly detection. The architecture is visualized in Fig. 6.

The activation function used in the RNN layers is tanh, which helps regulate gradient flow and prevents vanishing gradients in deep networks. The Nadam optimizer is employed to dynamically adjust the learning rate, ensuring faster convergence. The model is trained using categorical cross-entropy loss for multi-class classification and binary cross-entropy loss for anomaly detection. To enhance generalization, a batch size of 32 is used, and training runs for up to 200 epochs, with early stopping monitoring validation loss to prevent overfitting. Through this architecture, the RNN model effectively learns temporal patterns in IoT network traffic, achieving high accuracy in intrusion detection while maintaining efficiency in real-time threat analysis.

The common hyperparameters with their values are mentioned in Table 1.

3.4 Evaluation metrics

The evaluation metrics utilized in this research are shown in Table 2. These metrics provide a comprehensive assessment of the model's performance, ensuring an accurate and reliable evaluation of its effectiveness in detecting and classifying network intrusions.

4 Experimental results and analysis

This section presents the experimental results and analysis of the proposed model, evaluating its performance using various metrics on both multi-class classification and anomaly detection tasks.

The experimental setup for this research was implemented using Google Colab, leveraging its high-performance cloud-based GPU for efficient model training and evaluation. The implementation utilized key Python libraries, including TensorFlow and Keras for deep learning model development, Scikit-learn for data preprocessing and evaluation metrics, and Pandas and NumPy for data manipulation and numerical computations. The dataset was standardized using StandardScaler, and labels were encoded with LabelEncoder and converted into categorical format using to_categorical. The models, including 1D CNN, LSTM, MLP, and RNN, were constructed using Sequential API, with layers such as Conv1D, LSTM, SimpleRNN, Dense, Dropout, and MaxPooling1D. The Nadam optimizer was employed to optimize learning

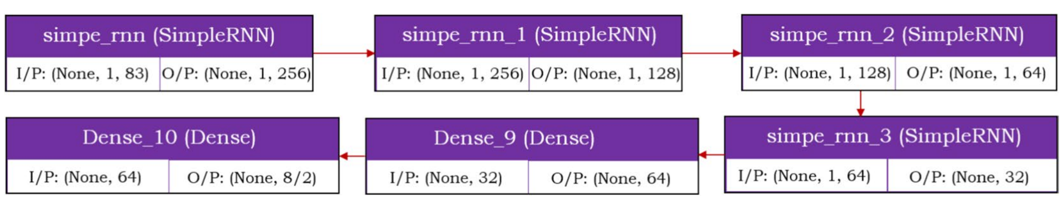

Fig. 6 Architecture of RNN

Table 1 Hyperparameters used in this research

Hyperparameter	Value	Description
Learning rate	0.001	Controls the step size in weight updates during training
Batch size	32	Number of samples processed before updating the model parameters
Epochs	200	Total number of iterations over the training dataset
Loss function	Categorical Crossentropy	Loss function for multi-class classification tasks
Optimizer	Nadam	Optimizer for model training using adaptive momentum estimation
Activation function	ReLU/Softmax	ReLU for hidden layers and Softmax for output layer
Dropout rate	0.5	Fraction of neurons randomly dropped to prevent overfitting
Kernel size	3	Size of the filter applied in convolutional layers
Pooling size	2	Size of the pooling window in max-pooling layers
Validation split	0.2	Fraction of data used for validation during training
Patience (early stopping)	10	Number of epochs without improvement before stopping training

Table 2 Evaluation metrics

Metric name	Equation	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Measures the proportion of correctly classified instances over total instances
Precision	$\frac{TP}{TP+FP}$	Indicates how many of the predicted positive instances were actually positive
Recall	$\frac{TP}{TP+FN}$	Represents the proportion of actual positives correctly identified
F1-score	$\frac{2*\text{Precision}*\text{Recall}}{\text{Precision}+\text{Recall}}$	Harmonic mean of precision and recall, balancing both metrics
Hamming loss	$\frac{1}{N}\sum_{i=1}^{N} I(y_i \neq \hat{y}_i)$	Measures the fraction of incorrect labels among all predicted labels
Classification report	N/A	Provides detailed performance analysis per class, including precision, recall, and F1-score
Confusion matrix	N/A	A matrix visualizing the number of correct and incorrect predictions across different classes

Table 3 Classification report for 1D CNN

	Precision	Recall	F1-score
Attack	0.99750	0.99405	0.99577
Benign	0.99255	0.99687	0.99471
Accuracy	0.99530		
Weighted Avg	0.99531	0.99530	0.99530

efficiency, while categorical cross-entropy loss was used for multi-class classification. EarlyStopping was integrated to prevent overfitting, ensuring optimal model performance. The training and validation performance was analyzed using classification reports, confusion matrices, ROC curves, calibration curves, and loss-accuracy graphs, providing a comprehensive evaluation of model effectiveness in network intrusion detection.

4.1 Results for anomaly detection

The classification report for the 1D CNN model, as presented in Table 3, demonstrates the model's strong performance in anomaly detection. The model achieves an overall accuracy of 99.53%, indicating a highly effective classification capability.

Figure 7 illustrates the accuracy and loss curves for the 1D CNN model during training and validation. The left plot depicts the training and validation accuracy, showing a steady increase as epochs progress, stabilizing at approximately 99.5%, indicating effective learning. The right plot presents the loss curves, where both training and validation losses decrease progressively, demonstrating proper convergence. The slight fluctuations in validation accuracy and loss suggest some variance, but the model maintains strong generalization. The results confirm that the 1D CNN effectively minimizes loss while maintaining high classification accuracy in anomaly detection.

Figure 8 presents the confusion matrix for the 1D CNN model in anomaly detection. The model correctly classifies 45,949 attack instances and 36,659 benign instances, with minimal misclassifications (275 benign misclassified as attack, 115 attack misclassified as benign). These results highlight the high classification accuracy and robustness of the model in distinguishing between attack and benign traffic.

Figure 9 illustrates the calibration curve for the 1D CNN model, comparing the predicted probabilities with actual outcomes for attack and benign traffic. The closer the curves align with the perfect calibration line (dashed line), the better the model's probability estimates. The figure indicates that the model's predictions are well-calibrated, demonstrating reliable confidence scores for classification.

Figure 10 presents the ROC Curve for the 1D CNN model in binary classification. The curve demonstrates a near-perfect AUC (0.99985), indicating an exceptionally

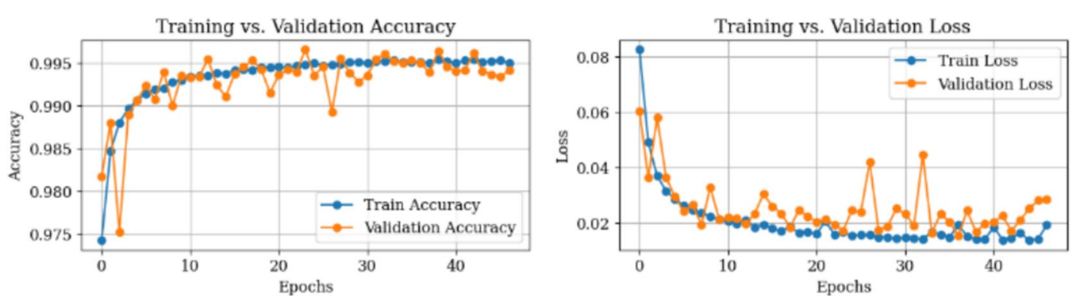

Fig. 7 Accuracy and loss of model for 1D CNN

high ability to distinguish between benign and attack traffic. The model achieves a high true positive rate with a minimal false positive rate, showcasing its effectiveness in anomaly detection.

Figure 11 illustrates the distribution of predicted probabilities for the 1D CNN model in anomaly detection. The histogram shows that the model confidently assigns probabilities close to 0 or 1, indicating strong classification confidence. The low overlap between benign (0) and attack (1) predictions confirms the model's effectiveness in distinguishing between normal and malicious traffic.

Table 4 presents the classification report for LSTM in anomaly detection. The model achieves a high accuracy of 99.52%, demonstrating its strong predictive capability. The precision, recall, and F1-score values for both attack and benign traffic exceed 99%, indicating balanced performance. The weighted average metrics further confirm the model's effectiveness in detecting anomalies with minimal misclassification.

Figure 12 illustrates the accuracy and loss curves for the LSTM model during training and validation. The training and validation accuracy curves show a steady improvement, stabilizing above 99%, indicating effective learning.

Figure 13 presents the confusion matrix for the LSTM model, illustrating its classification performance. The model correctly classifies 45,911 attack samples and 36,686 benign samples, with minimal misclassifications of 313 false positives and 88 false negatives.

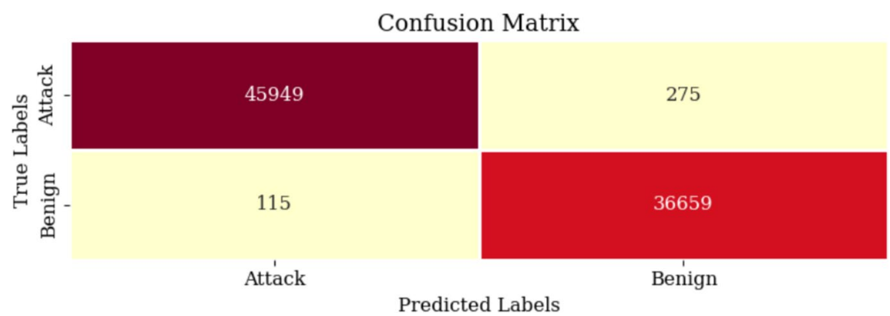

Fig. 8 Confusion matrix for 1D CNN

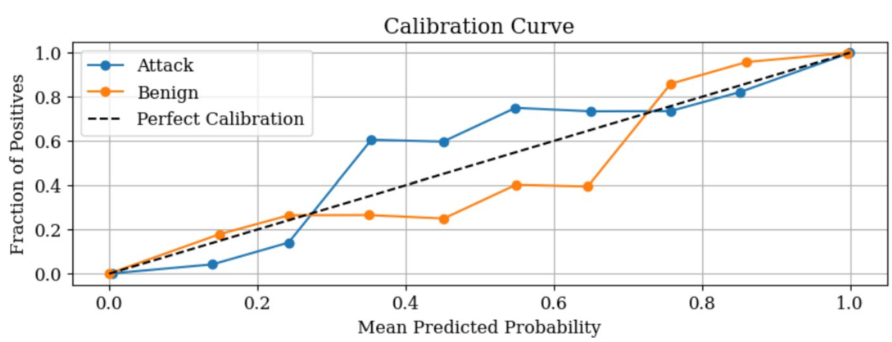

Fig. 9 Calibration curve for 1D CNN

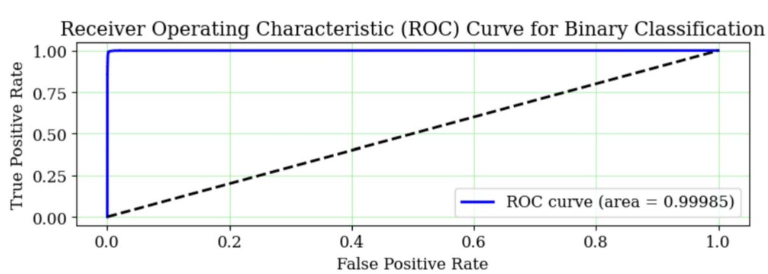

Fig. 10 ROC curve for 1D CNN

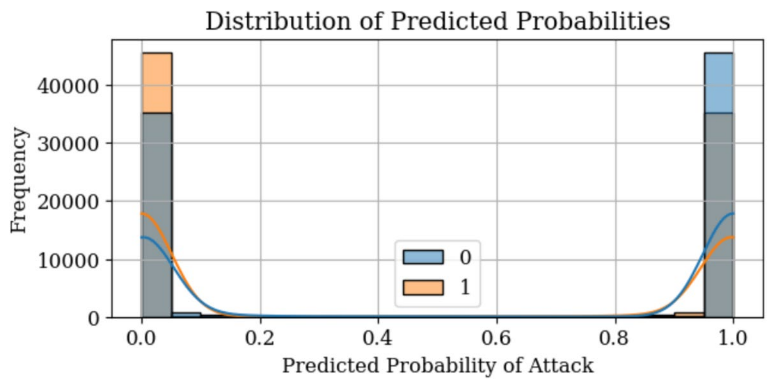

Fig. 11 Distribution of predicted probabilities for 1D CNN

Table 4 Classification report for LSTM

	Precision	Recall	F1-score
Attack	0.99809	0.99323	0.99565
Benign	0.99154	0.99761	0.99456
Accuracy	0.99517		
Weighted Avg	0.99519	0.99517	0.99517

Table 5 presents the classification report for the MLP model, demonstrating its effectiveness in anomaly detection. The model achieves an accuracy of 98.78%, with high precision (99.89%) for attacks and 97.45% for benign traffic. The recall values indicate strong detection capability, with 99.86% recall for benign samples and 97.92% for attack samples.

Figure 14 illustrates the accuracy and loss trends for the MLP model during training and validation. The accuracy curve shows a steady increase, stabilizing near 98.78%, indicating strong learning convergence. The loss graph demonstrates a consistent decline, with validation loss closely following training loss, confirming that the model generalizes well without overfitting.

Figure 15 presents the confusion matrix for the MLP model, showcasing its classification performance for attack and benign traffic. The model accurately

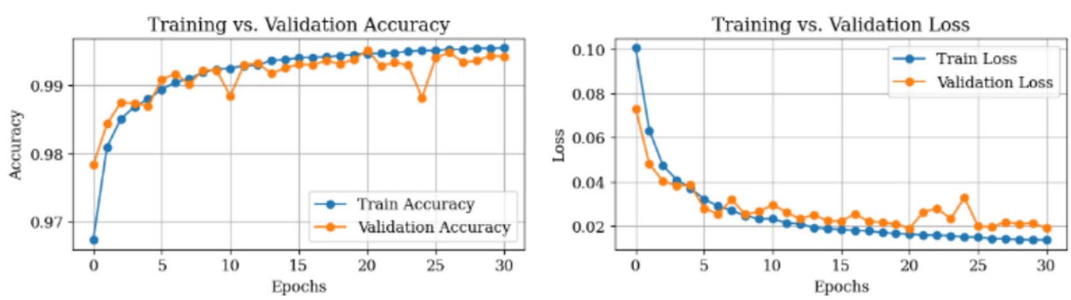

Fig. 12 Accuracy and loss of model for LSTM

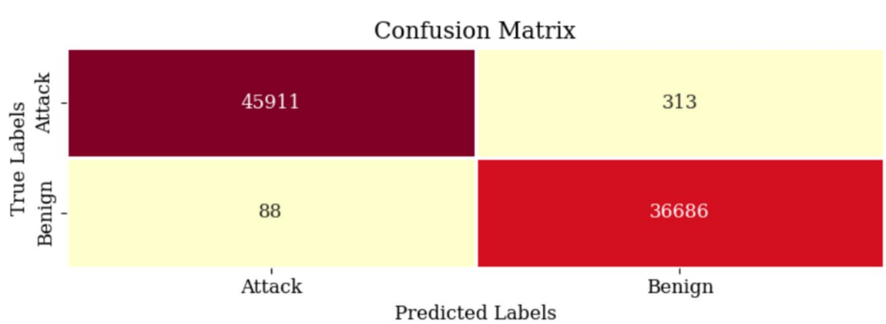

Fig. 13 Confusion matrix for LSTM

Table 5 Classification report for MLP

	Precision	Recall	F1-score
Attack	0.99890	0.97921	0.98896
Benign	0.97450	0.99864	0.98642
Accuracy	0.98782		
Weighted Avg	0.98809	0.98782	0.98783

Table 6 Classification report for RNN

	Precision	Recall	F1-score
Attack	0.99819	0.98830	0.99322
Benign	0.98547	0.99774	0.99157
Accuracy	0.99248		
Weighted Avg	0.99255	0.99248	0.99249

classifies 45,263 attack samples and 36,724 benign samples, with 961 false positives and 50 false negatives.

Table 6 presents the classification report for the RNN model, illustrating its effectiveness in detecting anomalies. The model achieves a high accuracy of 99.25%, with an F1-score of 0.99322 for attacks and 0.99157 for benign traffic.

Figure 16 presents the accuracy and loss curves for the RNN model. The left graph demonstrates a steady increase in training and validation accuracy, stabilizing around 99% after multiple epochs. The right graph shows a gradual decrease in both training and validation loss, indicating that the model learns effectively while avoiding significant overfitting.

Figure 17 illustrates the confusion matrix for the RNN model. The model correctly classifies 45,683 attack instances and 36,691 benign instances, with 541 false positives and 83 false negatives.

4.2 Results for multiclass classification

This section presents the performance evaluation of the models for multiclass classification.

Table 7 presents the classification report for the 1D CNN model on multiclass classification. The model achieves high precision, recall, and F1-scores across different attack categories, with an overall accuracy of 99.116%. The performance is particularly strong for common attack types like DoS, DDoS, and Spoofing, but slightly lower for XSS and SQL Injection. The weighted average metrics further confirm the model's robustness in handling multiclass scenarios.

Figure 18 illustrates the training and validation accuracy and loss curves for the 1D CNN model in multiclass classification. The accuracy graph shows a steady improvement, with the validation accuracy closely following the training accuracy, indicating minimal overfitting. The loss curve demonstrates a sharp decline in the early epochs, stabilizing at a lower value, confirming

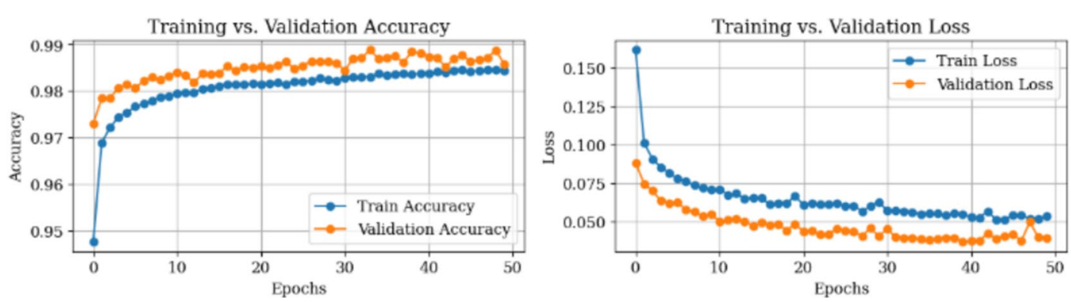

Fig. 14 Accuracy and loss of model for MLP

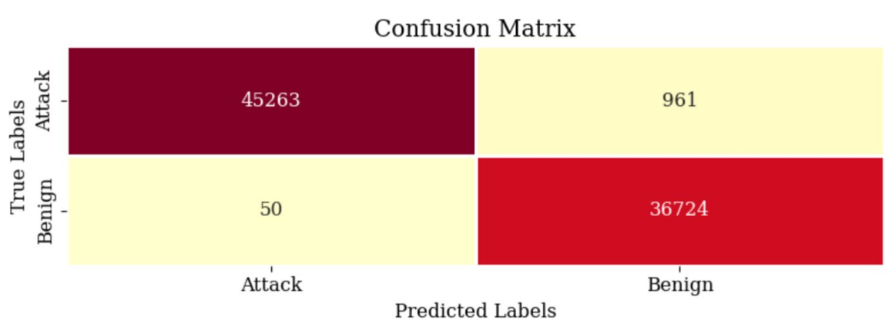

Fig. 15 Confusion matrix for MLP

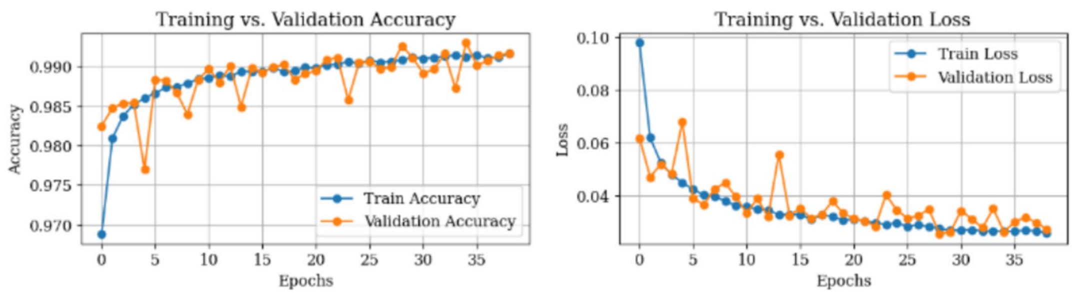

Fig. 16 Accuracy and loss of model for RNN

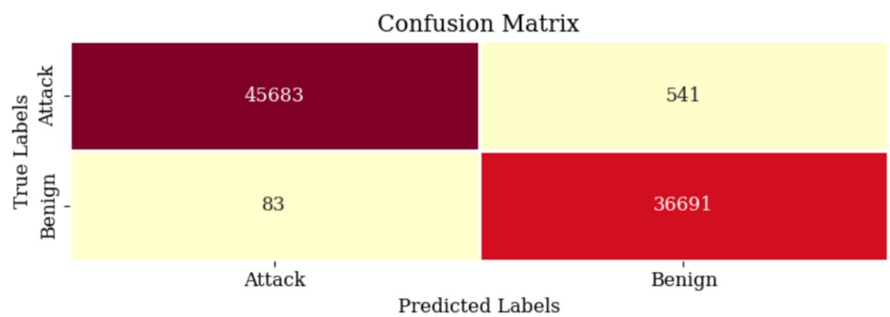

Fig. 17 Confusion matrix for RNN

Table 7 Classification report for 1D CNN

	Precision	Recall	F1-score
Benign	0.99291	0.99776	0.99533
BruteForce	0.91931	0.89481	0.90689
DDoS	0.99614	0.99398	0.99506
DoS	0.99730	0.99490	0.99610
Mirai	0.97511	0.84593	0.90594
Spoofing	0.99475	0.99333	0.99404
SqlInjection	0.95477	0.88694	0.91960
XSS	0.78272	0.96646	0.86494
Accuracy	0.99116		
Weighted Avg	0.99146	0.99116	0.99115

effective learning. The fluctuations in validation loss are minor, showing the model's stability in handling multi-class classification.

Figure 19 presents the confusion matrix for the 1D CNN model in multiclass classification, showcasing the model's classification performance across different attack types. The diagonal values represent correct classifications, with Benign (36,550), DDoS (10,073), DoS (17,743), and Spoofing (14,603) achieving strong recognition. Misclassifications are relatively low, indicating the model's high accuracy in distinguishing different attack categories. The matrix highlights the model's capability to handle complex network traffic data effectively.

Figure 20 illustrates the receiver operating characteristic (ROC) curve for the 1D CNN model in multiclass

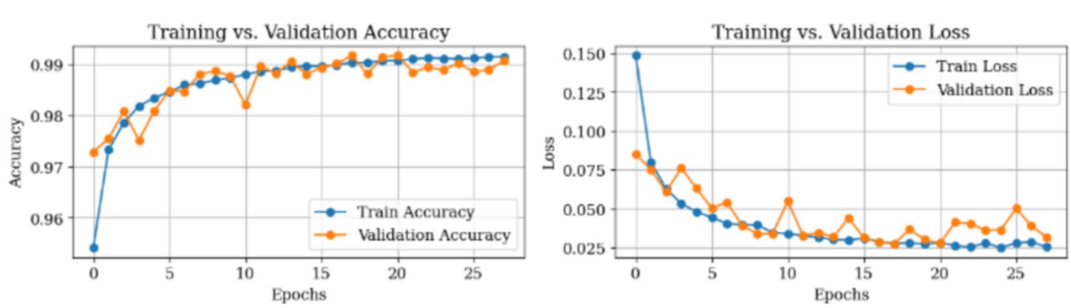

Fig. 18 Accuracy and loss of model for 1D CNN

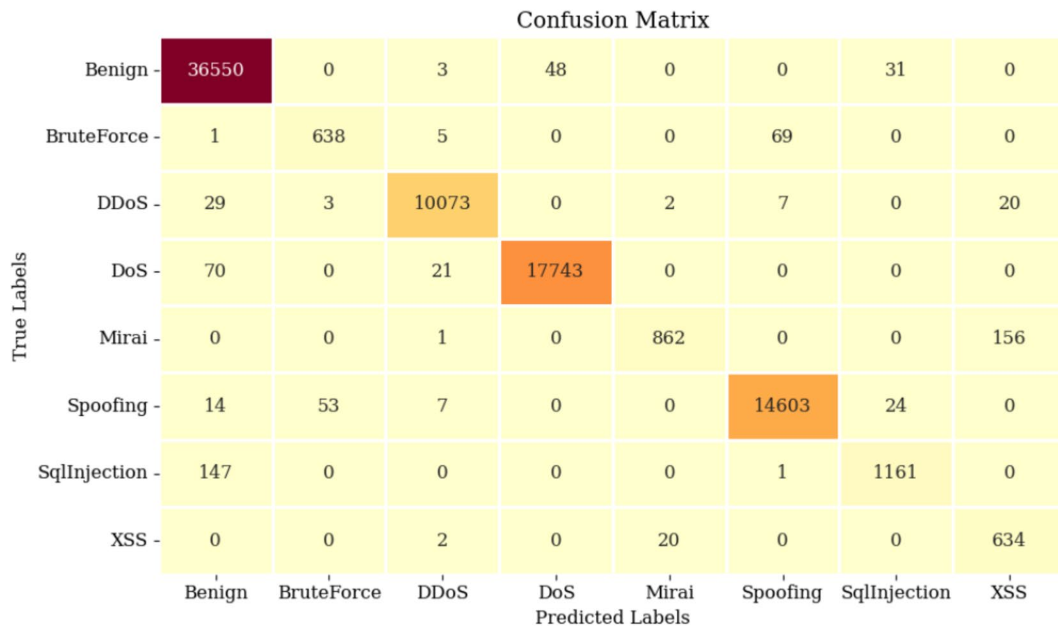

Fig. 19 Confusion matrix for 1D CNN

classification. Each class has an Area Under the Curve (AUC) score close to 1, indicating excellent discrimination ability across attack types. Notably, classes such as DDoS (0.99997), DoS (0.99996), and Benign (0.99976) demonstrate near-perfect classification performance. The near ideal ROC curve suggests that the model effectively minimizes false positives while achieving high true positive rates across all attack categories.

Figure 21 presents the calibration curve for the 1D CNN model in multiclass classification. The calibration curve compares the predicted probability of each class to the actual fraction of positives. While some classes align closely with the perfect calibration line (dashed line), others exhibit deviations, indicating slight miscalibrations in probability estimations. The model tends to overestimate or underestimate probabilities for certain attack categories, which may impact confidence scores in classification decisions.

Table 8 provides the classification report for the LSTM model in multiclass classification. The model achieves high precision, recall, and F1-score across most attack types, with an overall accuracy of 98.98%. While common attacks such as DoS and DDoS are detected with exceptional performance (F1-score >0.99), performance slightly declines for SqlInjection (F1-score =0.88402) and XSS (F1-score =0.89861), indicating some misclassification for these categories.

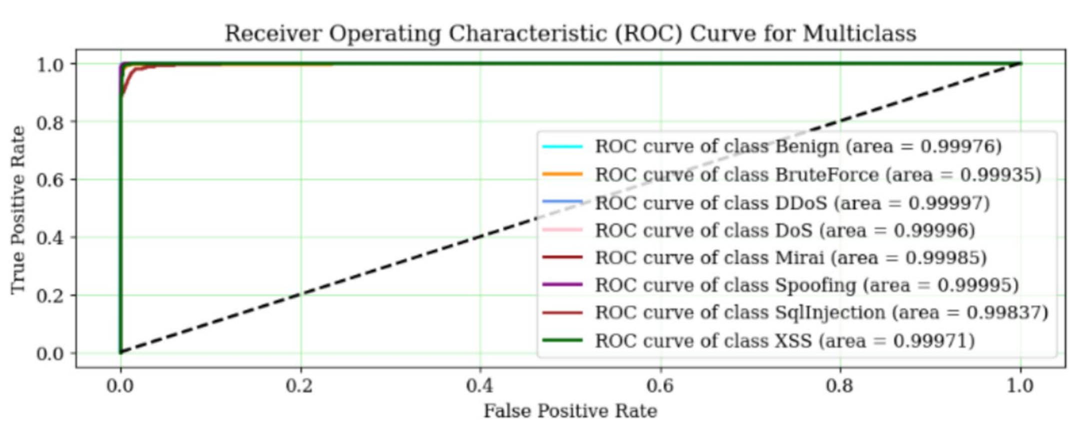

Fig. 20 ROC curve for 1D CNN

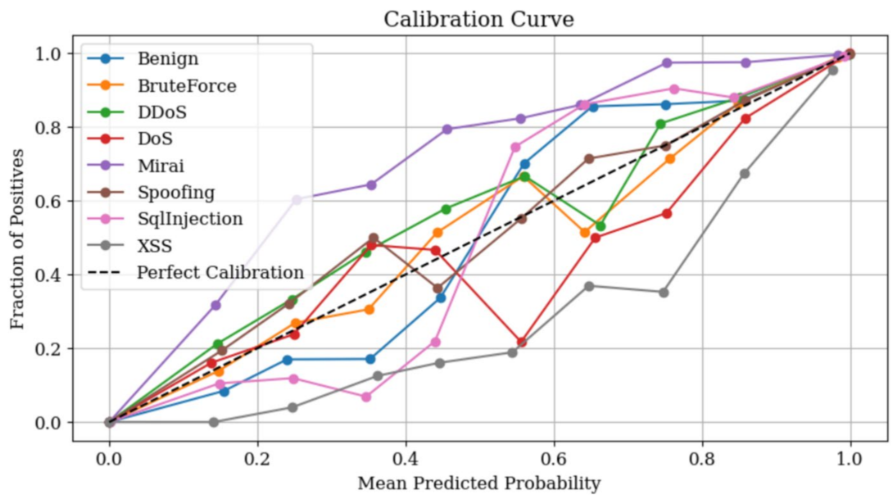

Fig. 21 Calibration curve for 1D CNN

Table 8 Classification report for LSTM

	Precision	Recall	F1-score
Benign	0.99036	0.99831	0.99432
BruteForce	0.84115	0.90603	0.87238
DDoS	0.99732	0.99122	0.99426
DoS	0.99781	0.99535	0.99658
Mirai	0.94903	0.91364	0.93100
Spoofing	0.99411	0.98653	0.99030
SqlInjection	0.94209	0.83270	0.88402
XSS	0.86154	0.93902	0.89861
Accuracy	0.98981		
Weighted Avg	0.98990	0.98981	0.98976

The weighted average scores suggest that the model maintains strong generalization across all classes.

Figure 22 presents the training vs. validation accuracy and loss curves for the LSTM model in multiclass classification. The accuracy plot shows a steady increase in both training and validation accuracy, eventually converging near 99%, demonstrating effective learning. The loss curve illustrates a rapid decline in both training and validation loss, stabilizing after around 10 epochs, indicating that the model successfully minimizes errors while preventing overfitting.

Figure 23 presents the confusion matrix for the LSTM model in multiclass classification. The diagonal values indicate the number of correctly classified samples for each class, showing strong performance across all categories. The highest misclassification occurs in attack types with similar characteristics, such as DDoS and DoS or Spoofing and BruteForce, which can be challenging to distinguish. The model exhibits high precision and recall for most classes, with minimal misclassifications.

Table 9 presents the classification report for the MLP model in multiclass classification. The model achieves a high accuracy of 97.2%, demonstrating its effectiveness in distinguishing different attack types. The Benign, DDoS, and DoS classes achieve the highest precision and recall scores, indicating strong performance in correctly classifying these instances. However, SqlInjection and XSS have relatively lower recall and F1-scores, suggesting

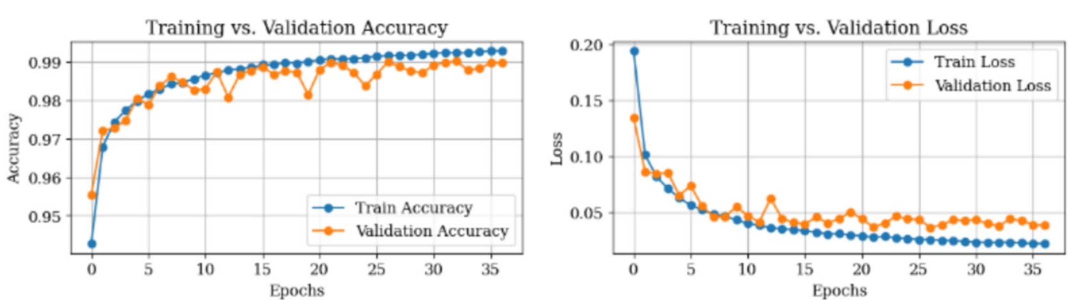

Fig. 22 Accuracy and loss of model for LSTM

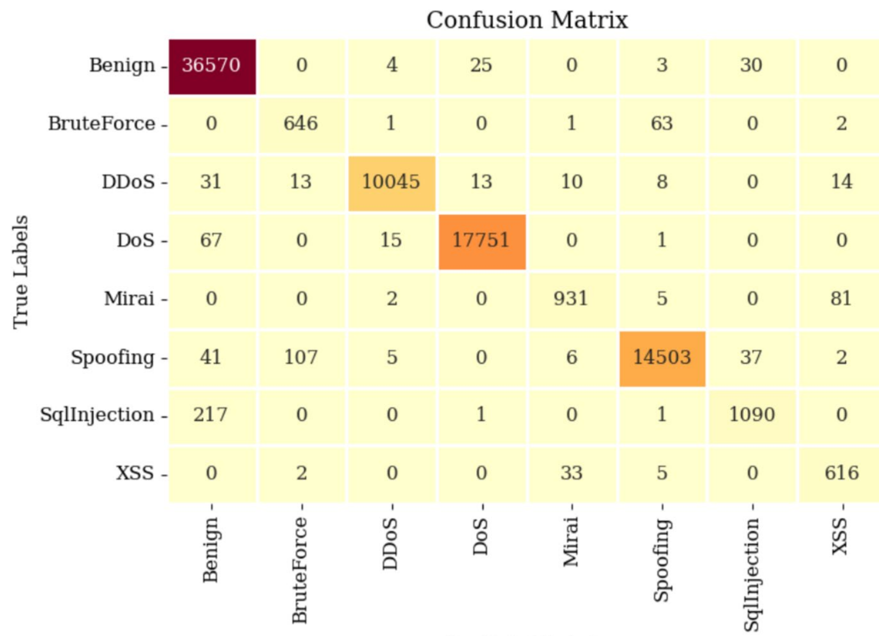

Fig. 23 Confusion matrix for LSTM

Table 9 Classification report for MLP

	Precision	Recall	F1-score
Benign	0.96395	0.99937	0.98134
BruteForce	0.81167	0.68303	0.74181
DDoS	0.99233	0.97010	0.98109
DoS	0.99869	0.98037	0.98945
Mirai	0.91837	0.79490	0.85218
Spoofing	0.97822	0.97456	0.97639
SqlInjection	0.89612	0.44156	0.59161
XSS	0.66387	0.84299	0.74278
Accuracy	0.97206		
Weighted Avg	0.97210	0.97206	0.97050

some misclassification, possibly due to similarities with other attacks.

Figure 24 illustrates the training vs. validation accuracy and training vs. validation loss for the MLP model in multiclass classification. The accuracy curve shows a steady improvement in both training and validation accuracy, reaching above 96% by the final epochs, indicating effective learning. The loss curve demonstrates a consistent decline, with validation loss stabilizing, suggesting that the model generalizes well to unseen data while avoiding overfitting.

Figure 25 presents the confusion matrix for the MLP model in multiclass classification. The diagonal values

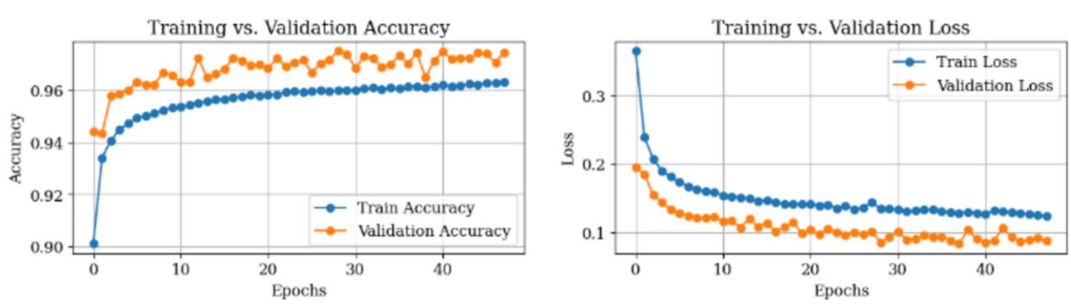

Fig. 24 Accuracy and loss of model for MLP

indicate the correct classifications for each class, with Benign, DoS, Spoofing, and DDoS achieving high correct classification rates. However, misclassifications are noticeable, particularly for SQL Injection and XSS attacks, suggesting that these attack types are harder to distinguish.

Table 10 presents the classification report for the RNN model in multiclass classification. The model achieves high precision, recall, and F1-score across most attack types, with DDoS, DoS, and Spoofing attacks showing the best performance. However, BruteForce, SQL Injection, and XSS attacks have lower recall values, indicating that the model struggles to correctly identify some instances of these attack types.

Figure 26 illustrates the training and validation accuracy (left) and training and validation loss (right) for the RNN model in multiclass classification. The training and validation accuracy curves indicate that the model converges well, reaching an accuracy close to 98%. The loss graph shows a rapid decrease in both training and validation loss, demonstrating effective learning. However, slight fluctuations in validation loss suggest minor instability, but the model maintains a strong generalization capability.

Figure 27 presents the confusion matrix for the RNN model in multiclass classification. The model shows strong classification performance, with a high number of correctly predicted instances for each class, particularly for Benign (36,513), DoS (17,688), and DDoS (10,013) attacks. However, some misclassifications are observed, especially for BruteForce and SQL Injection attacks, which might indicate overlapping feature patterns. The overall distribution highlights the model's robustness in detecting network intrusions.

Table 10 Classification report for RNN

	Precision	Recall	F1-score
Benign	0.98617	0.99675	0.99143
BruteForce	0.86349	0.73633	0.79485
DDoS	0.99523	0.98806	0.99163
DoS	0.99634	0.99181	0.99407
Mirai	0.93592	0.87439	0.90411
Spoofing	0.98306	0.98687	0.98496
SqlInjection	0.90119	0.75248	0.82015
XSS	0.76738	0.87500	0.81766
Accuracy	0.98432		
Weighted Avg	0.98417	0.98432	0.98404

4.3 Comparison of different metrics

Table 11 provides a performance comparison of different deep learning models, 1D CNN, LSTM, MLP, and RNN for anomaly detection. Among them, 1D CNN achieves the highest accuracy (99.53%), followed closely by LSTM (99.52%), while MLP (98.78%) and RNN (99.25%) show slightly lower performance. Precision, recall, and

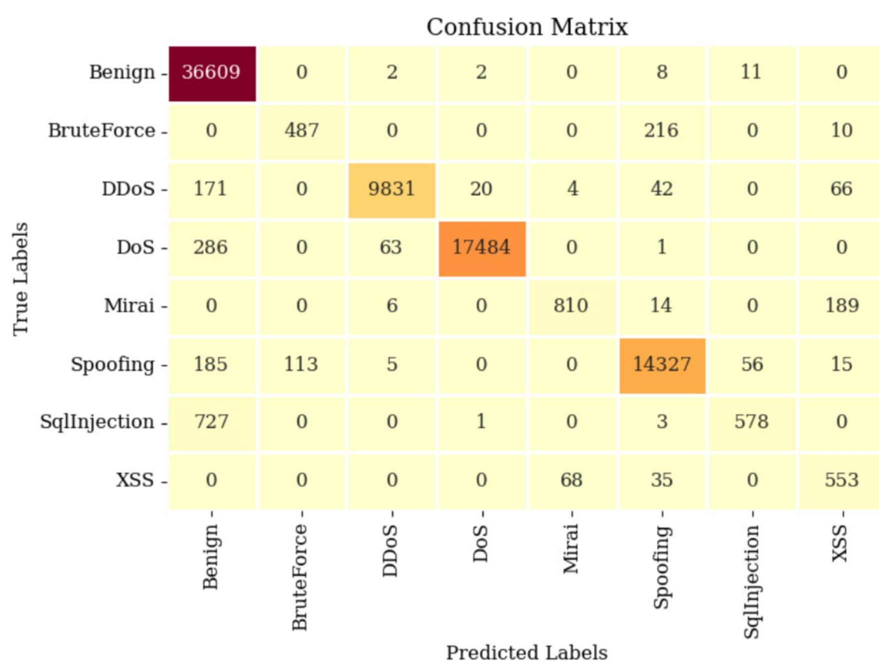

Fig. 25 Confusion matrix for MLP

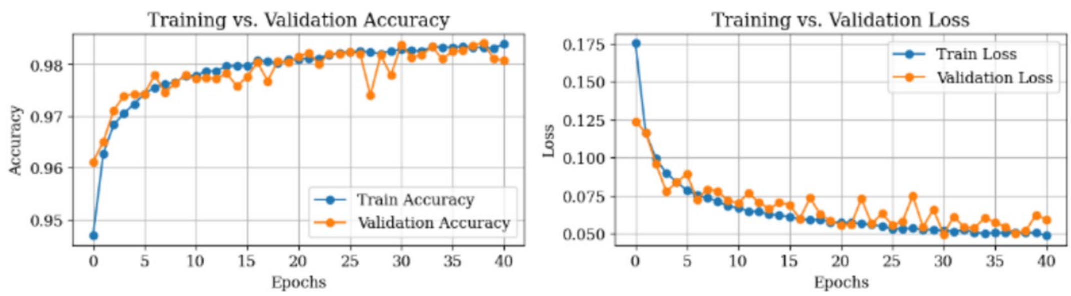

Fig. 26 Accuracy and loss of model for RNN

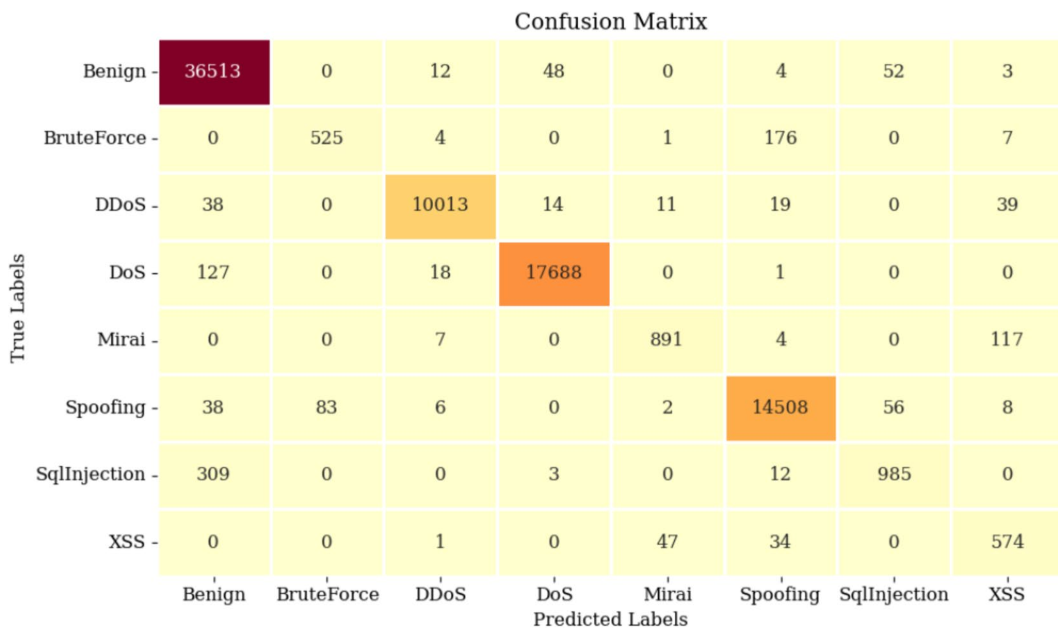

Fig. 27 Confusion matrix for RNN

Table 11 Comparison of different deep learning models for anomaly detection

Classifier	Accuracy	Precision	Recall	F1-score
1D CNN	0.99530	0.99531	0.99530	0.99530
LSTM	0.99517	0.99519	0.99517	0.99517
MLP	0.98782	0.98809	0.98782	0.98783
RNN	0.99248	0.99255	0.99248	0.99249

Table 12 Comparison of different deep learning models for multiclass classification

Classifier	Accuracy	Precision	Recall	F1-score
1D CNN	0.99116	0.99146	0.99116	0.99115
LSTM	0.98981	0.98990	0.98981	0.98976
MLP	0.97206	0.97210	0.97206	0.97050
RNN	0.98432	0.98417	0.98432	0.98404

F1-score values also reflect a consistent trend, reinforcing the superior effectiveness of CNN-based architectures in identifying anomalies in network traffic.

Table 12 presents the performance comparison of various deep learning models including 1D CNN, LSTM, MLP, and RNN for multiclass classification. The 1D CNN model achieves the highest accuracy (99.12%), demonstrating its superior generalization capability across multiple attack classes. LSTM follows closely with 98.98% accuracy, while RNN and MLP achieve 98.43% and 97.20%, respectively. Overall, CNN-based architectures outperform others, highlighting their robust feature extraction capability for complex multi-class scenarios.

Table 13 presents the evaluation results of the proposed 1D CNN model across multiple benchmark intrusion detection datasets, including CIC IoT-DIAD 2024, TabularIoTAttack-2024, CICIoT2023, N-BaIoT, and CSE-CIC-IDS2018. The model exhibits exceptional performance, achieving accuracy values consistently exceeding 98%, with precision, recall, and F1-scores all maintaining similarly high levels across different datasets. The lowest Hamming loss observed is 0.00004, demonstrating the model's minimal misclassification rates, reinforcing its robustness and reliability for both anomaly detection and multiclass classification. The superior results validate the efficacy of the proposed approach in real-world intrusion detection, where dynamic and sophisticated cyber threats pose significant security challenges. The success of the proposed 1D CNN model can be attributed to its ability to autonomously extract intricate spatial–temporal patterns in network traffic, eliminating the dependency on manual feature engineering while enhancing generalization across diverse attack scenarios. Unlike conventional machine learning methods that rely heavily on handcrafted features and struggle with high-dimensional data, the 1D CNN efficiently processes large-scale datasets, as demonstrated by its impressive performance on N-BaIoT (over 1 million records) and CSE-CIC-IDS2018 (over 2 million records) without compromising accuracy. Furthermore, its high recall scores indicate the model's effectiveness in detecting attack instances, significantly reducing the risk of false negatives which is an essential requirement in cybersecurity applications. The remarkably low Hamming loss across datasets underscores the model's reliability, ensuring minimal misclassification and reinforcing its robustness for real-world deployment. Collectively, these results establish the superiority of deep learning-driven intrusion detection over traditional approaches, positioning the proposed 1D CNN model as a scalable, adaptive, and high-performing solution for safeguarding IoT and network infrastructures against evolving cyber threats.

The proposed model demonstrates a low false positive rate (FPR) and a false negative rate (FNR) across multiple datasets. These indicate that the model effectively minimizes unnecessary alerts while maintaining a high detection capability.

Table 14 provides a comparative analysis of the proposed 1D CNN-based approach against existing deep learning models utilized in intrusion detection. The comparison highlights key performance metrics like accuracy, precision, recall, and F1-score for various architectures, including DNN, RNN, LSTM, CNN-LSTM, PCC-CNN, DL-BiLSTM, and Two-Stage LSTM. The results indicate that traditional deep learning approaches, such as DNN [29], RNN [43], and LSTM [44], exhibit relatively lower performance in terms of accuracy and recall, with some models missing critical metric values. The PCC-CNN model [45] achieves an

Table 14 Comparison of the proposed approach with existing approaches

Model	Accuracy	Precision	Recall	F1-score
DNN [29]	93.74	93.71	93.82	93.47
RNN [43]	*	96.52	96.52	71.40
EIDM [31]	95.00	*	*	*
PCC-CNN [45]	97.00	87.00	69.00	73.00
CNN + LSTM [46]	92.90	*	*	*
DIDS [47]	93.02	*	*	*
LSTM [44]	97.10	*	*	*
DNN [48]	90.79	89.41	81.64	85.34
FDLNN [49]	96.12	94.75	95.23	94.98
DL-BiLSTM [50]	98.61	66.52	73.18	67.23
CNN [51]	98.00	98.00	97.00	98.60
Two Stage-LSTM [52]	91.27	*	*	*
Proposed	99.00 +	99.00 +	99.00 +	99.00 +

(*) means not mentioned

Table 13 Performances of the proposed 1D CNN model for different datasets

Dataset	Classification	Data size (Amount × Features)	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Hamming loss
CIC IoT-DIAD 2024 [37]	8	414,990 rows × 84 columns	99.12	99.15	99.12	99.12	0.00884
	Anomaly		99.53	99.53	99.53	99.53	0.00470
TabularIoTAttack-2024 [39]	7	139,642 rows × 85 columns	99.30	99.31	99.30	99.30	0.00702
	Anomaly		99.92	99.92	99.92	99.92	0.00079
CICIoT2023 [40]	34	2,618,348 rows × 48 columns	98.37	98.26	98.37	98.21	0.00912
	Anomaly		99.30	99.40	99.30	99.33	0.00705
N-BaIoT [41]	Anomaly	1,018,298 rows × 116 columns	99.99	99.99	99.99	99.99	0.00007
CSE-CIC-IDS2018 [42]	5	2,085,299 rows × 80 columns	99.99	99.99	99.99	99.99	0.00004
	Anomaly		99.99	99.99	99.99	99.99	0.00005

accuracy of 97.00%, but its recall and F1-score remain below 73%, suggesting a trade-off between classification performance and generalizability.

More complex architectures, such as FDLNN [49] and DL-BiLSTM [40], attempt to enhance feature extraction capabilities; however, they suffer from significant precision loss (66.52% in DL-BiLSTM) while achieving moderate recall and F1-score values. The CNN-LSTM hybrid model [46] and Two-Stage LSTM [52] lack complete performance metrics, making direct comparisons challenging. While CNN [51] achieves 98.00% accuracy, it does not outperform the proposed 1D CNN in terms of robustness and generalization across different datasets, as observed in Table 13.

These findings solidify the effectiveness of the proposed approach, demonstrating its superior performance score when benchmarked against existing techniques. The model not only enhances intrusion detection but also ensures minimal false negatives and optimal classification performance across various attack scenarios. The ability to efficiently learn patterns from raw network traffic data, coupled with minimal computational overhead, positions the proposed 1D CNN as a state-of-the-art solution for intrusion detection, surpassing existing architectures in both performance and scalability.

In addition to evaluating accuracy metrics, we also analyzed the computational complexity and resource requirements of each model to assess their feasibility for real-time IoT applications. The 1D CNN model demonstrated the lowest computational overhead due to its efficient feature extraction and reduced reliance on sequential dependencies. LSTMs and RNNs, while effective in capturing temporal patterns, required significantly higher training time and memory consumption due to their recurrent structure. The MLP model, although computationally efficient, exhibited lower performance in feature extraction compared to 1D CNNs. These observations highlight that 1D CNNs offer an optimal trade-off between accuracy and computational efficiency, making them more suitable for real-time intrusion detection in IoT environments.

The proposed 1D CNN model demonstrates exceptional effectiveness in both anomaly detection and multiclass classification, achieving the highest accuracy, precision, recall, and F1-score among all evaluated models. Its ability to efficiently capture spatial dependencies in sequential data makes it highly robust for network intrusion detection. The model consistently outperforms LSTM, MLP, and RNN, proving its superiority in feature extraction and classification performance while maintaining computational efficiency.

5 Conclusion

In this research, a 1D Convolutional Neural Network (1D CNN)-based approach was developed for intrusion detection in IoT and network security environments, addressing the growing need for robust and intelligent threat detection mechanisms. The study explored multiple deep learning models, including LSTM, RNN, and MLP, evaluating their effectiveness for both anomaly detection and multiclass classification. Experimental results demonstrated that the proposed 1D CNN model consistently outperformed other architectures, achieving superior accuracy, recall, and F1-score across multiple benchmark datasets. Comparative analysis with existing approaches further highlighted the model's efficacy in detecting sophisticated cyberattacks with minimal false positives and hamming loss, reinforcing its adaptability and precision in real-world cybersecurity applications. The significance of this work lies in its ability to leverage lightweight convolutional operations to efficiently process large-scale network traffic data while maintaining high detection rates and computational efficiency. Ultimately, this research establishes a state-of-the-art deep learning framework for intrusion detection, paving the way for further advancements in automated cybersecurity solutions and real-time threat intelligence in IoT and cloud-based environments.

5.1 Integration into IoT security framework

The proposed 1D CNN-based intrusion detection system plays a crucial role within a layered IoT security framework, complementing other security measures such as encryption, access control, and anomaly-based monitoring. In a defense-in-depth strategy, this intrusion detection system (IDS) functions as an active monitoring layer, continuously analyzing network traffic to detect and mitigate potential threats in real time. By integrating the IDS with existing network firewalls and security information and event management (SIEM) systems, organizations can enhance threat intelligence by correlating detection outputs with broader security data. Furthermore, incorporating this IDS into edge computing environments allows real-time threat mitigation with minimal latency, making it highly suitable for resource-constrained IoT networks. The ability to perform deep packet inspection and automated feature extraction enables the model to identify sophisticated attack patterns with greater accuracy compared to traditional signature-based detection systems. As cyber threats evolve, the IDS can serve as a proactive security mechanism within zero-trust architectures, where continuous authentication and anomaly detection are critical. This integration ensures that IoT ecosystems remain secure against both known and emerging threats.

5.2 Deployment challenges and future considerations

While the proposed 1D CNN-based intrusion detection system demonstrates high accuracy and computational efficiency, real-world deployment presents several challenges. Latency is a critical factor in IoT environments, as intrusion detection systems must process large volumes of network traffic in real-time. Although 1D CNNs are computationally efficient, optimizing inference time for large-scale deployments remains an area of improvement. Model interpretability is another challenge, as deep learning models function as black-box systems, making it difficult to understand decision-making processes. Future work will focus on integrating explainable AI (XAI) techniques to enhance transparency. Additionally, adaptability to evolving threats is essential, as cyber threats in IoT networks continuously evolve. Adversarial robustness must be improved to counter evolving attack techniques, ensuring that intrusion detection systems remain resilient to sophisticated cyber threats. Furthermore, while the proposed IDS is effective within the context of a single dataset, its generalization across heterogeneous IoT environments with diverse protocols and communication architectures remains an area for further exploration. The future focus should be on developing models capable of handling variations in attack patterns across different IoT ecosystems. Also, integrating federated learning can enhance privacy and scalability, allowing distributed IoT devices to collaboratively learn attack patterns without exposing sensitive data. Finally, exploring self-learning adaptive models will enable IDS to dynamically evolve against emerging threats, improving their long-term effectiveness in real-world IoT ecosystems.

Authors' contributions
Md. Alamgir Hossain: Design, implement and writing the complete paper.

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Data availability
The datasets utilized in this research are publicly available and have been cited in the dataset "Dataset" section.

Declarations

Competing interests
The author declares no competing interests.

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